Geometry-Aware JEPA for EEG

A controlled frozen-transfer study on TUAB

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Hack the World(s) – EEG Track · Team **HelloWorlds**

Contribution

Our contribution, in one slide

1. **A strong frozen baseline.** SIGReg-ambient JEPA matches the LeJEPA/Laya recipe's value on our setup – not a controlled head-to-head.
2. **Nothing beats it.** Ambient/tangent SIGReg & PEIRA, Graph-JEPA, Fourier-JEPA – none beats ambient SIGReg in balanced accuracy.
3. **Latent geometry is the contribution.** Visualised with **de Surrel's AIRM Riemannian t-SNE** (the right metric for SPD/EEG covariances).

Motivation

The question we actually asked

- **EEG foundation models** are everywhere – but most report *fine-tuned* numbers. What survives when you **freeze** the encoder?
- **JEPA** avoids collapse with an *anti-collapse regulariser*. The EEG covariance lives on a curved **SPD manifold**, not in flat Euclidean space.
- **Our intuition**: a *Riemannian geometry-aware* anti-collapse regulariser – acting in the SPD **tangent** of the EEG covariance, not flat ambient space – should *respect and improve* the latent geometry.
- **What we ship**: a strong frozen probe *and* a geometry-aware regulariser that **measurably sharpens the latent-space geometry**, visualised with de Surrel's AIRM Riemannian t-SNE.

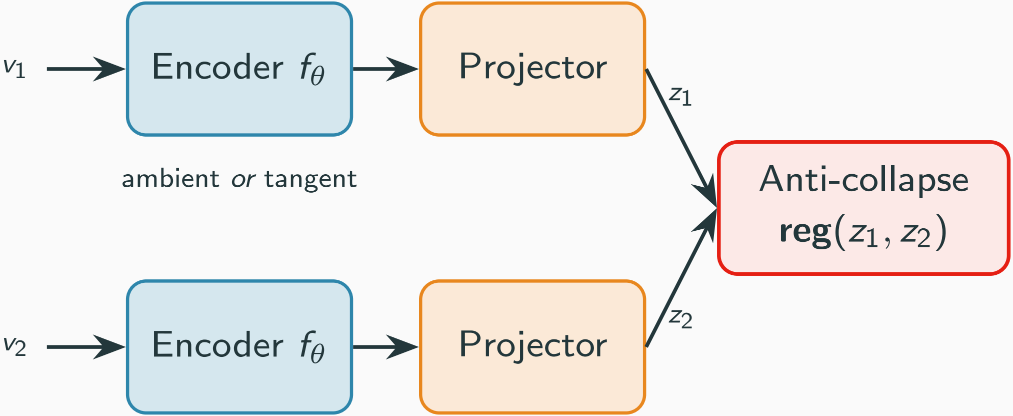
Data

Data: TUH Abnormal EEG (TUAB)

- **Task**: binary *normal* vs *abnormal* EEG, recording level.
- **Signal**: 19 channels @ 200 Hz; 10 s windows (2,000 samples).
- **Standard split**: 2,717 train / 276 eval recordings, **patient-disjoint**.
- **Pretraining**: SSL reads only *unlabeled* train recordings.

Architecture

Two view and 1D convolutional encoder



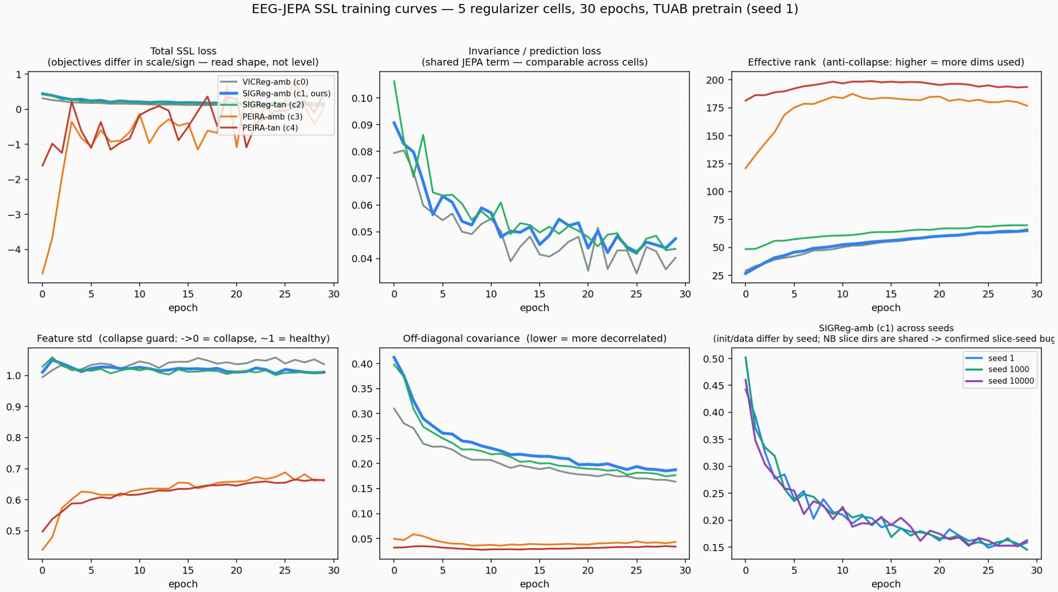
What we tried

regulariser \ **space**	**ambient** (Euclidean)	**tangent** (SPD)
VICReg (variance/cov)	`C0` reference	—
SIGReg (Gaussianity slices)	`C1` *Laya-like baseline*	`C2` geometry-aware
PEIRA (distribution-free)	`C3`	`C4` *ex-hypothesis*

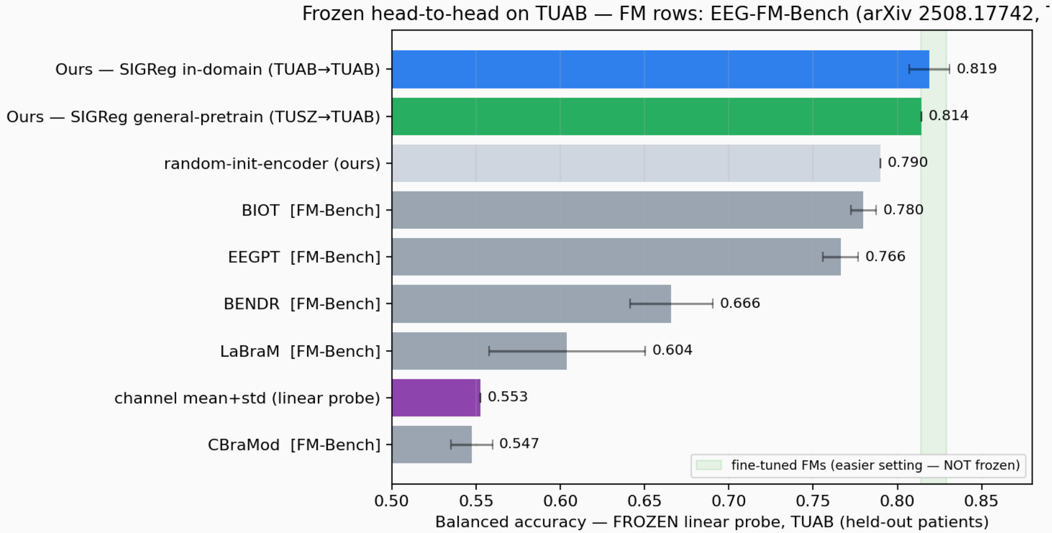
- **SIGReg** (LeJEPA/BCS): tests random 1-D slices for Gaussianity – assumes an *isotropic* target.
- **PEIRA**: distribution-free anti-collapse – no target distribution to mis-specify.
- **GraphJEPA**
- **FourierJEPA**

Results

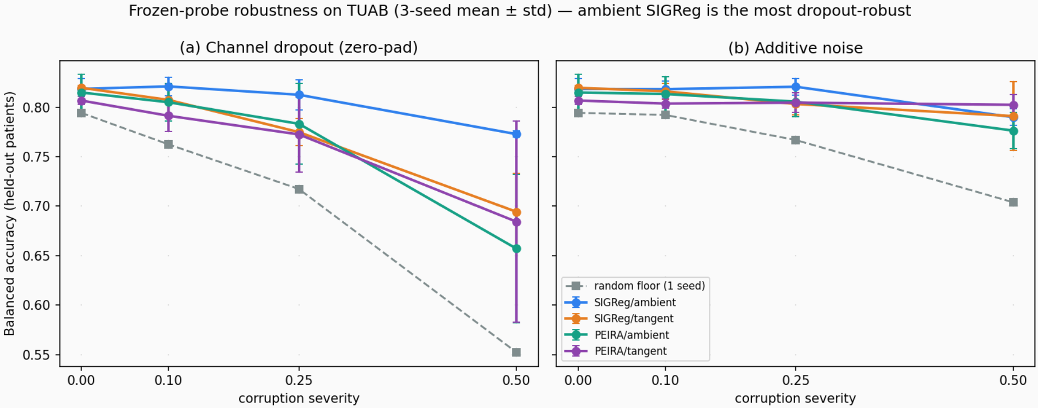
SSL training curves (all cells healthy)



Frozen & in-domain: we hold where frozen FMs fall

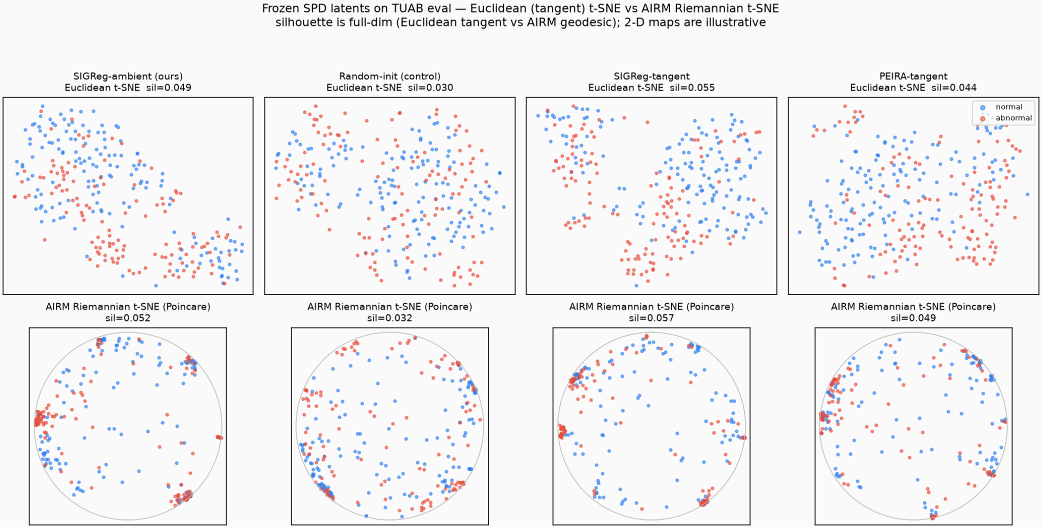


Beyond accuracy: frozen-probe robustness (3-seed)



Visualization

Manifold-aware latent geometry (AIRM vs Euclidean)



Manifold viz generalises across sustained TUH tasks

Conclusion

Conclusion

- A **controlled frozen-transfer study**, not a leaderboard entry.
- **A strong frozen probe** (0.819 BA, $\sim$0.89 AUROC) that holds where published FMs collapse when frozen.
- **The payoff is the latent geometry**: a Riemannian geometry-aware (SPD-tangent) regulariser **sharpens the frozen latent** – normal/abnormal and seizure classes form coherent structure – and **de Surrel's AIRM Riemannian embedding sharpens it further over the Euclidean view**, most clearly for SIGReg.
- On this simple binary task the probe accuracy is the same across variants – the tangent's gain is *geometric*, exactly where a manifold-aware regulariser should help.

Built on a trimmed `eb_jepa` vendor.

References

References i

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References iv

- `eb_jepa` – JEPA reference implementation, Meta AI / FAIR. github.com/facebookresearch/eb_jepa. *(our trimmed code vendor)*